

Software-Defined Optics Architecture for Device Independent Learning-Based Optical Control

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ABSTRACT

Modern reconfigurable optical systems are often built around tightly integrated, device-specific stacks, linking sensing, decision-making, and actuation to a particular combination of detectors, mirrors, lenses, modulators, or phased arrays. Replacing any of these components can therefore require substantial redesign of the high-level control application. This work proposes a three-layer Software-Defined Optics (SDO) architecture that separates optical intelligence from the underlying hardware. The central objective is application portability: the same training and inference software can be retained when the sensing or optical hardware changes, while adaptation is confined to lower-layer device models, capability descriptors, drivers, and configuration parameters. The architecture is demonstrated through a common optical-control API that allows learning models to be exchanged while preserving the underlying control stack and optical system. By separating optical applications from physical devices, the proposed architecture enables reusable control software, modular hardware replacement, and rapid deployment of learning-based optical functions across heterogeneous platforms.

Keywords: Software-Defined Optics, Photonic Instrumentation Architecture.